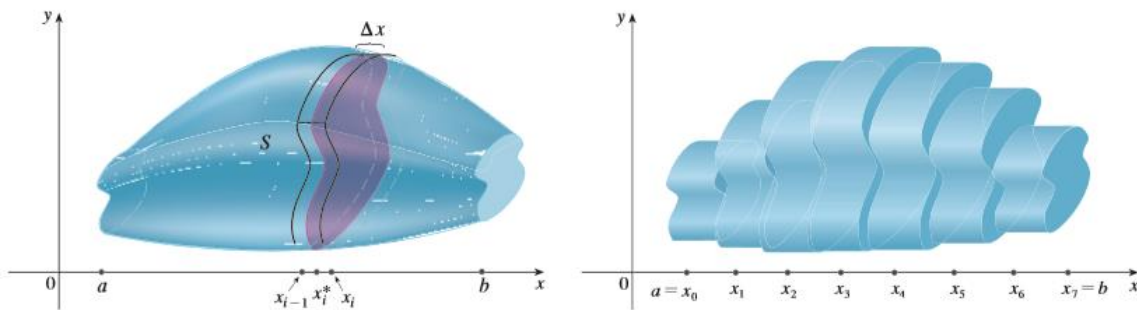


Applications of Integrals (Ch. 5)

Inverse Functions (Ch. 6)

5.2 Volumes



Definition of Volume Let S be a solid that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane P_x , through x and perpendicular to the x -axis, is $A(x)$, where A is a continuous function, then the **volume** of S is

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i^*) \Delta x = \int_a^b A(x) dx$$

In general, we calculate the volume of a solid of revolution by using the basic defining formula

$$V = \int_a^b A(x) dx \quad \text{or} \quad V = \int_c^d A(y) dy$$

and we find the cross-sectional area $A(x)$ or $A(y)$ in one of the following ways:

- If the cross-section is a disk, we find the radius of the disk (in terms of x or y) and use

$$A = \pi(\text{radius})^2$$

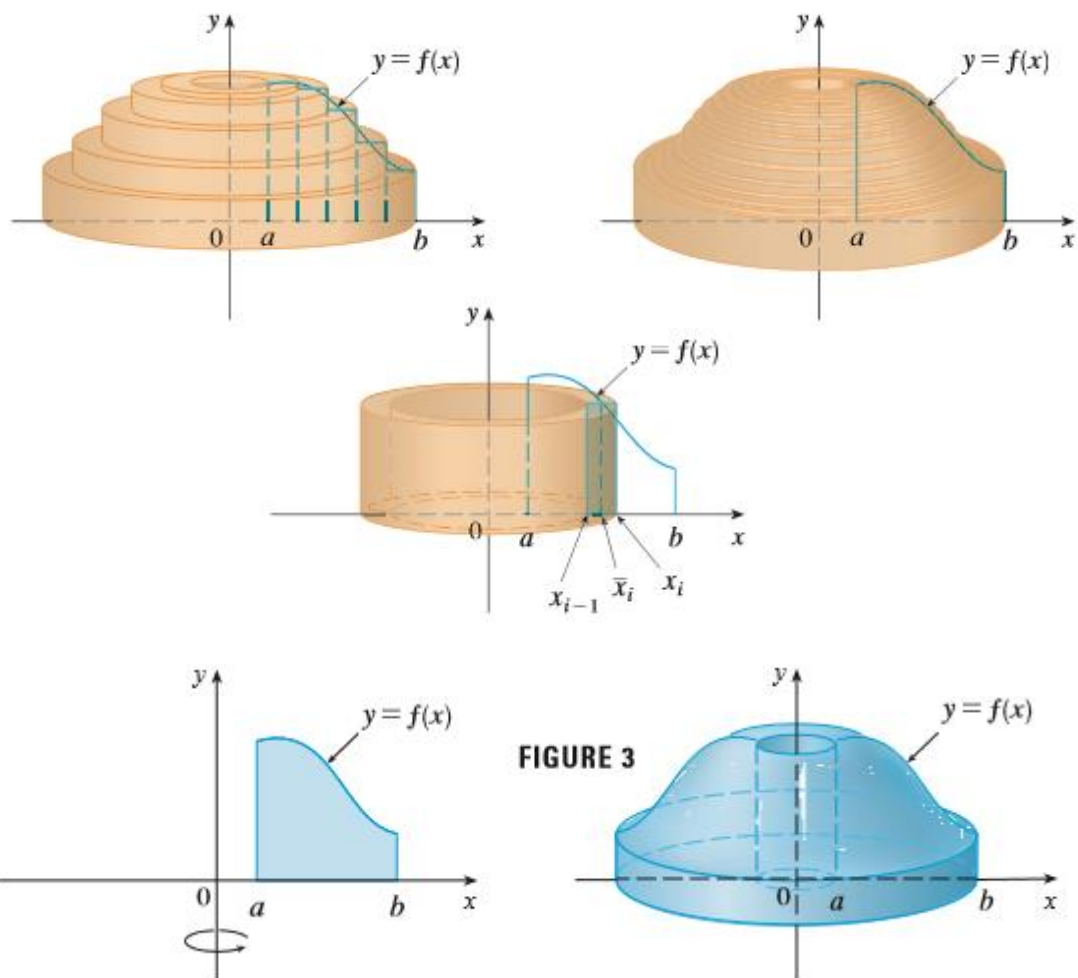
- If the cross-section is a washer (as in Examples 4 and 5), we find the inner radius r_{in} and outer radius r_{out} from a sketch (as in Figure 10) and compute the area of the washer by subtracting the area of the inner disk from the area of the outer disk:

$$A = \pi(\text{outer radius})^2 - \pi(\text{inner radius})^2$$

FIGURE 10



5.3 Volumes by Cylindrical Shells



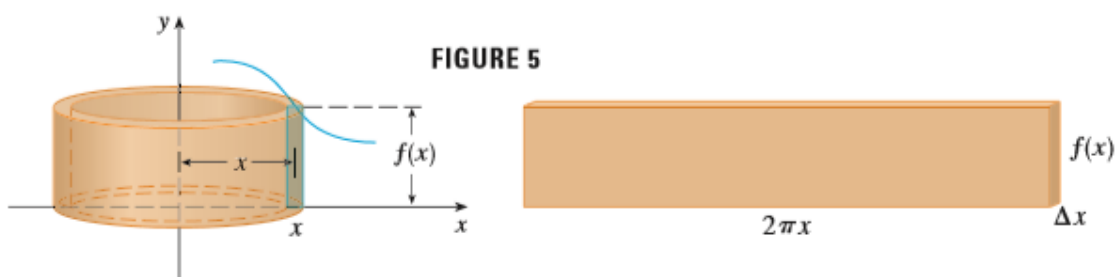
The volume of the solid in Figure 3, obtained by rotating about the y -axis the region under the curve $y = f(x)$ from a to b , is

$$V = \int_a^b 2\pi x f(x) dx \quad \text{where } 0 \leq a < b$$

The argument using cylindrical shells makes Formula 2 seem reasonable, but later we will be able to prove it (see Exercise 71 in Section 7.1).

The best way to remember Formula 2 is to think of a typical shell, cut and flattened as in Figure 5, with radius x , circumference $2\pi x$, height $f(x)$, and thickness Δx or dx :

$$\int_a^b \underbrace{(2\pi x)}_{\text{circumference}} \underbrace{[f(x)]}_{\text{height}} \underbrace{dx}_{\text{thickness}}$$



Chapter 6 – first half(next)

6.1 Inverse Functions

Definition A function f is called a **one-to-one function** if it never takes on the same value twice; that is,

$$f(x_1) \neq f(x_2) \quad \text{whenever } x_1 \neq x_2$$

Horizontal Line Test A function is one-to-one if and only if no horizontal line intersects its graph more than once.

Definition Let f be a one-to-one function with domain A and range B . Then its **inverse function** f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \iff f(x) = y$$

for any y in B .

$$\text{domain of } f^{-1} = \text{range of } f$$

$$\text{range of } f^{-1} = \text{domain of } f$$

⊗ **CAUTION** Do not mistake the -1 in f^{-1} for an exponent. Thus

$$f^{-1}(x) \text{ does not mean } \frac{1}{f(x)}$$

For domain-and-range being denoted by A -and- B for $f(x)$ we have

$$f^{-1}(x) = y \iff f(y) = x$$

cancellation equations:

$$f^{-1}(f(x)) = x \text{ for every } x \text{ in } A$$

$$f(f^{-1}(x)) = x \text{ for every } x \text{ in } B$$

How to Find the Inverse Function of a One-to-One Function f

Step 1 Write $y = f(x)$.

Step 2 Solve this equation for x in terms of y (if possible).

Step 3 To express f^{-1} as a function of x , interchange x and y .
The resulting equation is $y = f^{-1}(x)$.

The graph of f^{-1} is obtained by reflecting the graph of f about the line $y = x$.

Fact: If the inverse function of $f(\cdot)$ exists and $f'(f^{-1}(a)) \neq 0$ then

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

6.2* Natural Logarithmic function

(Please note that Section 6.2 is different from this 'star' Section 6.2*)

Natural Logarithms

The logarithm with base e is called the **natural logarithm** and has a special notation:

$$\log_e x = \ln x$$

$$\ln x = y \iff e^y = x$$

$$\ln(e^x) = x \quad x \in \mathbb{R}$$

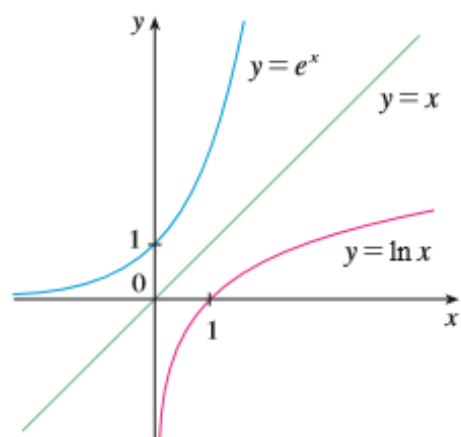
$$e^{\ln x} = x \quad x > 0$$

In particular, if we set $x = 1$, we get

$$\ln e = 1$$

Change of Base Formula For any positive number a ($a \neq 1$), we have

$$\log_a x = \frac{\ln x}{\ln a}$$



The graph of $y = \ln x$ is the reflection of the graph of $y = e^x$ about the line $y = x$

Logarithmic Functions

If $a > 0$ and $a \neq 1$, the exponential function $f(x) = a^x$ is either increasing or decreasing and so it is one-to-one by the Horizontal Line Test. It therefore has an inverse function f^{-1} , which is called the **logarithmic function with base a** and is denoted by $\log_a$.

$$\log_a x = y \iff a^y = x$$

$$\log_a(a^x) = x \quad \text{for every } x \in \mathbb{R}$$

$$a^{\log_a x} = x \quad \text{for every } x > 0$$

Laws of Logarithms If x and y are positive numbers, then

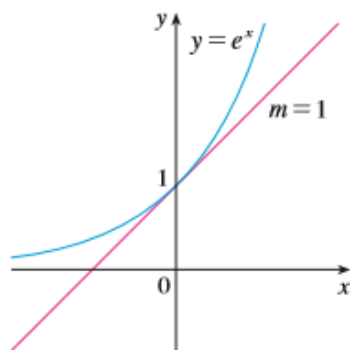
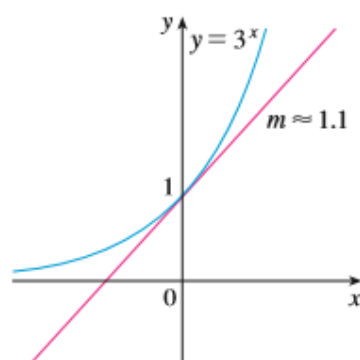
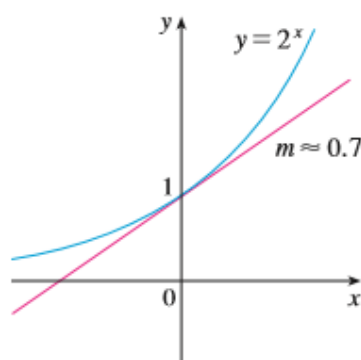
$$1. \log_a(xy) = \log_a x + \log_a y$$

$$2. \log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$3. \log_a(x^r) = r \log_a x \quad (\text{where } r \text{ is any real number})$$

The Number e

Of all possible bases for an exponential function, there is one that is most convenient for the purposes of calculus. The choice of a base a is influenced by the way the graph of $y = a^x$ crosses the y -axis. Next Figures show the tangent lines to the graphs of $y = 2^x$ and $y = 3^x$ at the point $(0, 1)$. (Tangent lines will be defined precisely in Section 2.7. For present purposes, you can think of the tangent line to an exponential graph at a point as the line that touches the graph only at that point.) If we measure the slopes of these tangent lines at $(0, 1)$, we find that $m \approx 0.7$ for $y = 2^x$ and $m \approx 1.1$ for $y = 3^x$.



It turns out, as we will see in Chapter 3, that some of the formulas of calculus will be greatly simplified if we choose the base a so that the slope of the tangent line to $y = a^x$ at $(0, 1)$ is *exactly* 1.

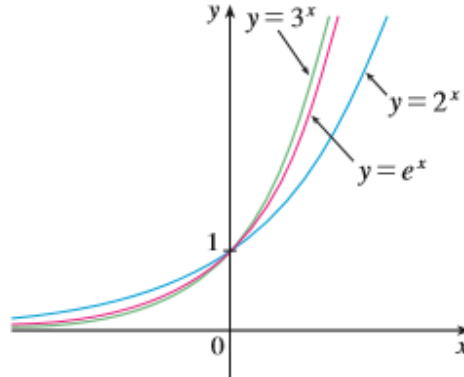
In fact, there *is* such a number and it is denoted by the letter e . (This notation was chosen by the Swiss mathematician Leonhard Euler in 1727, probably because it is the first letter of the word *exponential*.)

It comes as no surprise that the number e lies between 2 and 3 and the graph of $y = e^x$ lies between the graphs of $y = 2^x$ and $y = 3^x$.

In Chapter 3 we will see that the value of e , correct to five decimal places, is

$$e \approx 2.71828$$

We call the function $f(x) = e^x$ the **natural exponential function**.



Integral and derivative of $\ln(x)$... and limits as x goes to infinity or to zero-plus

Derivatives of Logarithmic Functions

$$\frac{d}{dx} (\log_a x) = \frac{1}{x \ln a}$$

$$\frac{d}{dx} (\ln x) = \frac{1}{x}$$

$$\frac{d}{dx} (\ln u) = \frac{1}{u} \frac{du}{dx}$$

or

$$\frac{d}{dx} [\ln g(x)] = \frac{g'(x)}{g(x)}$$

Logarithmic differentiation (shown/ done)

6.3* Natural Exponential function

(Please note that Section 6.3 is different from this 'star' Section 6.3*)

Relation to natural logarithm function.

Limits as x goes to plus/ minus infinity

(Derivative and integral)

Laws of Exponents If a and b are positive numbers and x and y are any real numbers, then

$$1. a^{x+y} = a^x a^y \quad 2. a^{x-y} = \frac{a^x}{a^y} \quad 3. (a^x)^y = a^{xy} \quad 4. (ab)^x = a^x b^x$$

the Chain Rule gives

$$\begin{aligned} \frac{d}{dx} (a^x) &= \frac{d}{dx} (e^{(\ln a)x}) = e^{(\ln a)x} \frac{d}{dx} (\ln a)x \\ &= e^{(\ln a)x} \cdot \ln a = \underline{a^x \ln a} \end{aligned}$$

6.4* General Logarithmic and Exponential function

(Please note that Section 6.4 is different from this 'star' Section 6.4*)

As seen above